

Chapter One

Derivation of a Partial Differential Equation

1.1 Partial Differential Equations

A differential equation involving partial derivatives of one or more dependent variables with respect to more than one independent variables is called a partial differential equation.

In general, z is considered as dependent variable and x and y are considered as independent variables.

The general form of partial differential equation is $\phi\left(x, y, z, \frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}\right) = 0$ which can be also written as

$\phi(x, y, z, p, q) = 0$, where p and q represents the partial differential coefficients $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ respectively.

Examples:

- (i) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$ is a partial differential equation, where u is dependent variable and x, y, z are independent variables.
- (ii) $\frac{\partial v}{\partial s} + \frac{\partial v}{\partial t} = v$ is a partial differential equation, where v is dependent variable and s, t are independent variables.

1.2 Order and Degree of a Partial Differential Education

Order: The order of the highest ordered derivative involved in a partial differential equation is called the order of the partial differential equation.

Degree: The degree of the highest order differential of an partial differential equation is called the degree of the partial differential equation, where differential equation is polynomial differential equation of differential coefficient.

Examples: (i) $\frac{\partial^4 z}{\partial x^4} + \left(\frac{\partial z}{\partial y}\right)^2 = z$; 4th order and 1st degree.

(ii) $\left(\frac{\partial^2 z}{\partial x^2}\right)^3 + x^4 \left(\frac{\partial z}{\partial y}\right)^3 = 2$; 2nd order and 3rd degree.

(iii) $x^3 p + y^2 q = 2z$; 1st order and 1st degree.

(iv) $p^2 + q^2 = m$; 1st order and 2nd degree.

1.3 Linear and Non-linear Partial Differential Equation

The equation will be called linear if, it is of first degree in the dependent variable and its partial derivatives (i.e. powers or products of the dependent variable and its partial derivatives must be absent). A partial differential equation which is not linear is called a non-linear partial differential equation.

Examples: (i) $z = x^2 p + y^2 q$ is a linear partial differential equation.

(ii) $p^2 + q^2 = 4$ is a non-linear partial differential equation.

1.4 Derivation of a Partial Differential Equation by the Elimination of Arbitrary Constants

Let us consider an equation $\varphi(x, y, z, a, b) = 0$ (1)

where a, b are arbitrary constants and z be regarded as function of two independent variables x and y .

Differentiating equation (1) with respect to x and y partially respectively, we get

$$\frac{\partial \varphi}{\partial x} + p \frac{\partial \varphi}{\partial z} = 0 \quad \dots (2)$$

$$\text{and } \frac{\partial \varphi}{\partial y} + q \frac{\partial \varphi}{\partial z} = 0 \quad \dots (3)$$

Eliminating two arbitrary constants a and b from equations (1), (2) and (3), we shall obtain an equation of the form $f(x, y, z, p, q) = 0$, which is a partial differential equation of order one.

In a similar manner, it can be shown that if there are more arbitrary constants than the number of independent variables, the above procedure of elimination will give rise to partial differential equations of higher order than the first.

Example-1 Form partial differential equation by eliminating arbitrary constants from the following equations:

$$(i) \quad z = (x + a)(y + b)$$

$$(ii) \quad z = ax + by + a^2 + b^2$$

Solution

$$(i) \text{ Given equation } z = (x + a)(y + b) \quad \dots (1)$$

Differentiating (1) partially w. r. to x and y respectively, we get

$$\frac{\partial z}{\partial x} = (y + b) \text{ and } \frac{\partial z}{\partial y} = (x + a)$$

$$\Rightarrow y + b = p \text{ and } x + a = q; \text{ since } p = \frac{\partial z}{\partial x}, q = \frac{\partial z}{\partial y}$$

Substituting these values in equation (1), we obtain

$$z = pq$$

which is the required partial differential equation.

$$(ii) \text{ Given equation } z = ax + by + a^2 + b^2 \quad \dots (1)$$

Differentiating (1) partially w. r. to x and y respectively, we get

$$\frac{\partial z}{\partial x} = a \text{ and } \frac{\partial z}{\partial y} = b$$

$$\Rightarrow a = p \text{ and } b = q; \text{ since } p = \frac{\partial z}{\partial x}, q = \frac{\partial z}{\partial y}$$

Substituting these values in equation (1), we obtain the required partial differential equation

$$z = px + qy + p^2 + q^2 \text{ (Ans.)}$$

1.5 Derivation of Partial Differential Equation by the Elimination of Arbitrary Function φ from the equation $\varphi(u, v) = 0$, where u and v are functions of x, y and z

$$\text{Given } \varphi(u, v) = 0 \quad \dots (1)$$

We consider z as dependent variable and x and y as independent variables

So, $\frac{\partial z}{\partial x} = p, \frac{\partial z}{\partial y} = q, \frac{\partial y}{\partial x} = 0$

Differentiating (1), partially with respect to x, we get

$$\frac{\partial \phi}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial \phi}{\partial v} \frac{\partial v}{\partial x} = 0$$

$$\text{or, } \frac{\partial \phi}{\partial u} \left(\frac{\partial u}{\partial x} \frac{\partial x}{\partial x} + \frac{\partial u}{\partial y} \frac{\partial y}{\partial x} + \frac{\partial u}{\partial z} \frac{\partial z}{\partial x} \right) + \frac{\partial \phi}{\partial v} \left(\frac{\partial v}{\partial x} \frac{\partial x}{\partial x} + \frac{\partial v}{\partial y} \frac{\partial y}{\partial x} + \frac{\partial v}{\partial z} \frac{\partial z}{\partial x} \right) = 0 \quad \dots (2)$$

$$\text{or, } \frac{\partial \phi}{\partial u} \left(\frac{\partial u}{\partial x} + p \frac{\partial u}{\partial z} \right) + \frac{\partial \phi}{\partial v} \left(\frac{\partial v}{\partial x} + p \frac{\partial v}{\partial z} \right) = 0$$

$$\text{or, } \frac{\partial \phi}{\partial u} / \frac{\partial \phi}{\partial v} = - \left(\frac{\partial v}{\partial x} + p \frac{\partial v}{\partial z} \right) / \left(\frac{\partial u}{\partial x} + p \frac{\partial u}{\partial z} \right) \quad \dots (3)$$

Similarly, differentiating (1), partially with respect to y, we get

$$\frac{\partial \phi}{\partial u} \left(\frac{\partial u}{\partial y} + q \frac{\partial u}{\partial z} \right) + \frac{\partial \phi}{\partial v} \left(\frac{\partial v}{\partial y} + q \frac{\partial v}{\partial z} \right) = 0$$

$$\text{or, } \frac{\partial \phi}{\partial u} / \frac{\partial \phi}{\partial v} = - \left(\frac{\partial v}{\partial y} + q \frac{\partial v}{\partial z} \right) / \left(\frac{\partial u}{\partial y} + q \frac{\partial u}{\partial z} \right) \quad \dots (4)$$

Eliminating ϕ from the equations (3) and (4), we get

$$\left(\frac{\partial u}{\partial x} + p \frac{\partial u}{\partial z} \right) \left(\frac{\partial v}{\partial y} + q \frac{\partial v}{\partial z} \right) - \left(\frac{\partial u}{\partial y} + q \frac{\partial u}{\partial z} \right) \left(\frac{\partial v}{\partial x} + p \frac{\partial v}{\partial z} \right) = 0$$

$$\Rightarrow \left(\frac{\partial u}{\partial z} \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial v}{\partial z} \right) p + \left(\frac{\partial u}{\partial x} \frac{\partial v}{\partial z} - \frac{\partial u}{\partial z} \frac{\partial v}{\partial x} \right) q = \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} - \frac{\partial u}{\partial x} \frac{\partial v}{\partial y}$$

$$\Rightarrow Pp + Qq = R \quad \dots (5)$$

where $P = \frac{\partial u}{\partial z} \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial v}{\partial z} = \begin{vmatrix} \frac{\partial u}{\partial z} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial z} & \frac{\partial v}{\partial y} \end{vmatrix}$

$Q = \frac{\partial u}{\partial x} \frac{\partial v}{\partial z} - \frac{\partial u}{\partial z} \frac{\partial v}{\partial x} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial z} \end{vmatrix}$ and $R = \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} - \frac{\partial u}{\partial x} \frac{\partial v}{\partial y} = \begin{vmatrix} \frac{\partial u}{\partial y} & \frac{\partial u}{\partial x} \\ \frac{\partial v}{\partial y} & \frac{\partial v}{\partial x} \end{vmatrix}$

Equation (5) is partial differential equation of order one and first degree known as Lagrange's linear partial differential equation.

13.5.1 THEOREM (Chain Rules for Derivatives) If $x = x(t)$ and $y = y(t)$ are differentiable at t , and if $z = f(x, y)$ is differentiable at the point $(x, y) = (x(t), y(t))$, then $z = f(x(t), y(t))$ is differentiable at t and

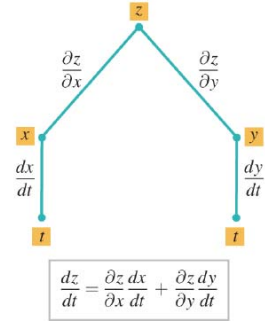
$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt} \quad (5)$$

where the ordinary derivatives are evaluated at t and the partial derivatives are evaluated at (x, y) .

If each of the functions $x = x(t)$, $y = y(t)$, and $z = z(t)$ is differentiable at t , and if $w = f(x, y, z)$ is differentiable at the point $(x, y, z) = (x(t), y(t), z(t))$, then the function $w = f(x(t), y(t), z(t))$ is differentiable at t and

$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt} + \frac{\partial w}{\partial z} \frac{dz}{dt} \quad (6)$$

where the ordinary derivatives are evaluated at t and the partial derivatives are evaluated at (x, y, z) .



Example-2 Form a partial differential equation by eliminating the arbitrary function ϕ from $\phi(x + y + z, x^2 + y^2 - z^2) = 0$. What is the order of this partial differential equation? [DUH-2006]

Solution Given $\phi(x + y + z, x^2 + y^2 - z^2) = 0$ (1)

Let, $u = x + y + z$ and $v = x^2 + y^2 - z^2$

$$\therefore \frac{\partial u}{\partial x} = 1, \frac{\partial u}{\partial y} = 1, \frac{\partial u}{\partial z} = 1, \frac{\partial v}{\partial x} = 2x, \frac{\partial v}{\partial y} = 2y, \frac{\partial v}{\partial z} = -2z$$

Equation (1) becomes $\phi(u, v) = 0$... (2)

Differentiating (2) partially w. r. to x , we get

$$\begin{aligned} & \frac{\partial \phi}{\partial u} \left(\frac{\partial u}{\partial x} \frac{\partial x}{\partial x} + \frac{\partial u}{\partial y} \frac{\partial y}{\partial x} + \frac{\partial u}{\partial z} \frac{\partial z}{\partial x} \right) + \frac{\partial \phi}{\partial v} \left(\frac{\partial v}{\partial x} \frac{\partial x}{\partial x} + \frac{\partial v}{\partial y} \frac{\partial y}{\partial x} + \frac{\partial v}{\partial z} \frac{\partial z}{\partial x} \right) = 0 \\ \Rightarrow & \frac{\partial \phi}{\partial u} (1 + 0 + p) + \frac{\partial \phi}{\partial v} (2x + 0 - 2zp) = 0 \quad \left[\because \frac{\partial y}{\partial x} = 0 \right] \\ \Rightarrow & \frac{\partial \phi / \partial u}{\partial \phi / \partial v} = \frac{2(zp - x)}{1 + p} \quad \dots (3) \end{aligned}$$

Again, differentiating (2) partially w. r. to y , we get

$$\begin{aligned} & \frac{\partial \phi}{\partial u} \left(\frac{\partial u}{\partial x} \frac{\partial x}{\partial y} + \frac{\partial u}{\partial y} \frac{\partial y}{\partial y} + \frac{\partial u}{\partial z} \frac{\partial z}{\partial y} \right) + \frac{\partial \phi}{\partial v} \left(\frac{\partial v}{\partial x} \frac{\partial x}{\partial y} + \frac{\partial v}{\partial y} \frac{\partial y}{\partial y} + \frac{\partial v}{\partial z} \frac{\partial z}{\partial y} \right) = 0 \\ \Rightarrow & \frac{\partial \phi}{\partial u} (1 + 0 + q) + \frac{\partial \phi}{\partial v} (2y + 0 - 2zq) = 0 \\ \Rightarrow & \frac{\partial \phi / \partial u}{\partial \phi / \partial v} = \frac{2(zq - y)}{1 + q} \quad \dots (4) \end{aligned}$$

Eliminating ϕ from the equations (3) and (4), we obtain

$$\begin{aligned} & \frac{2(zp - x)}{1 + p} = \frac{2(zq - y)}{1 + q} \\ \Rightarrow & (1 + q)(zp - x) = (1 + p)(zq - y) \end{aligned}$$

$$\Rightarrow (y+z)p - (x+z)q = x - y$$

which is the required partial differential equation of order one.

Alternative method :

$$\text{Given } \phi(x+y+z, x^2+y^2-z^2) = 0 \quad \dots (1)$$

$$\text{Let, } u = x+y+z \text{ and } v = x^2+y^2-z^2$$

$$\therefore \frac{\partial u}{\partial x} = 1, \frac{\partial u}{\partial y} = 1, \frac{\partial u}{\partial z} = 1, \frac{\partial v}{\partial x} = 2x, \frac{\partial v}{\partial y} = 2y, \frac{\partial v}{\partial z} = -2z$$

$$\text{Equation (1) becomes } \phi(u, v) = 0 \quad \dots (2)$$

We know, the partial differential equation of (2) is

$$Pp + Qq = R \quad \dots (3)$$

$$\text{where } P = \begin{vmatrix} \frac{\partial u}{\partial z} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial z} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ -2z & 2y \end{vmatrix} = 2(y+z)$$

$$Q = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial z} \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 2x & -2z \end{vmatrix} = -2(x+z)$$

$$R = \begin{vmatrix} \frac{\partial u}{\partial y} & \frac{\partial u}{\partial x} \\ \frac{\partial v}{\partial y} & \frac{\partial v}{\partial x} \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 2y & 2x \end{vmatrix} = 2(x-y)$$

Putting the values of P , Q and R in (3), we obtain the required partial differential equation

$$2(y+z)p - 2(x+z)q = 2(x-y)$$

$$\Rightarrow (y+z)p - (x+z)q = x - y$$

which is a first order partial differential equation.

Example-3 Form partial differential equations by eliminating arbitrary function from the following relations:

$$(i) \quad z = e^{ax+by} f(ax-by)$$

$$(ii) \quad (x+y+z) = f(x^2+y^2+z^2)$$

$$(iii) \quad z = e^{ay} f(x+by)$$

$$\text{Solution (i) Given } z = e^{ax+by} \cdot f(ax-by) \quad \dots (1)$$

Differentiating (1) partially w. r. to x and y respectively, we get

$$p = ae^{ax+by} \cdot f(ax-by) + ae^{ax+by} \cdot f'(ax-by) \quad \dots (2)$$

$$\text{and } q = be^{ax+by} \cdot f(ax-by) - be^{ax+by} \cdot f'(ax-by) \quad \dots (3)$$

$$(2)b + (3)a \Rightarrow bp + aq = 2ab e^{ax+by} \cdot f(ax-by)$$

$$\Rightarrow bp + aq = 2abz$$

which is the required partial differential equation.

$$(ii) \text{ Given, } x+y+z = f(x^2+y^2+z^2) \quad \dots (1)$$

Differentiating (1) partially w. r. to x and y respectively, we get

$$1 + \frac{\partial z}{\partial x} = f'(x^2 + y^2 + z^2) \cdot \left(2x + 2z \frac{\partial z}{\partial x} \right)$$

$$\Rightarrow 1 + p = 2(x + zp) f'(x^2 + y^2 + z^2) \quad \dots (2)$$

$$\text{and } 1 + \frac{\partial z}{\partial y} = f'(x^2 + y^2 + z^2) \cdot \left(2y + 2z \frac{\partial z}{\partial y} \right)$$

$$\Rightarrow 1 + q = 2(y + zq) \cdot f'(x^2 + y^2 + z^2) \quad \dots (3)$$

Dividing (2) by (3), we get

$$\frac{1+p}{1+q} = \frac{x+zp}{y+zq}$$

$$\Rightarrow (1+p)(y+zq) = (1+q)(x+zp)$$

$$\Rightarrow (y-z)p + (z-x)q = x-y$$

which is the required partial differential equation.

$$\text{(iii) Given } z = e^{ay} f(x+by) \quad \dots (1)$$

Differentiating (1) partially w. r. to x and y respectively, we get

$$p = e^{ay} \cdot f'(x+by) \quad \dots (2)$$

$$\text{and } q = ae^{ay} \cdot f(x+by) + be^{ay} \cdot f'(x+by) \quad \dots (3)$$

Using (1) and (2) in equation (3), we get

$$q = az + bp$$

$$\Rightarrow z = \frac{1}{a}(q - bp)$$

which is the required partial differential equation.

Example-4 Form partial differential equations by eliminating arbitrary constants from the following equations.

$$\text{(i) } z = ae^{bt} \sin bx$$

$$\text{(ii) } \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

Solution

$$\text{(i) Given } z = ae^{bt} \sin bx \quad \dots (1)$$

Differentiating (1) partially w. r. to x and t respectively, we get

$$\frac{\partial z}{\partial x} = abe^{bt} \cos bx \quad \dots (2)$$

$$\text{and } \frac{\partial z}{\partial t} = abe^{bt} \sin bx \quad \dots (3)$$

Again, differentiating (2) and (3) partially w. r. to x and t respectively, we get

$$\frac{\partial^2 z}{\partial x^2} = -ab^2 e^{bt} \sin bx \quad \dots (4)$$

$$\text{and } \frac{\partial^2 z}{\partial t^2} = ab^2 e^{bt} \sin bx \quad \dots (5)$$

Adding (4) & (5), we obtain the required partial differential equation

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0 \quad (\text{Ans.})$$

(ii) Given $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$... (1)

Differentiating (1) partially w. r. to x and y respectively, we get

$$\frac{2x}{a^2} + \frac{2z}{c^2} \frac{\partial z}{\partial x} = 0 \Rightarrow \frac{x}{a^2} + \frac{z}{c^2} \frac{\partial z}{\partial x} = 0 \quad \dots (2)$$

and $\frac{2y}{b^2} + \frac{2z}{c^2} \frac{\partial z}{\partial y} = 0 \Rightarrow \frac{y}{b^2} + \frac{z}{c^2} \frac{\partial z}{\partial y} = 0 \quad \dots (3)$

Again differentiating (2) and (3) w. r. to x and y respectively, we get

$$\frac{1}{a^2} + \frac{1}{c^2} \left(\frac{\partial z}{\partial x} \right)^2 + \frac{z}{c^2} \frac{\partial^2 z}{\partial x^2} = 0 \quad \dots (4)$$

and $\frac{1}{b^2} + \frac{1}{c^2} \left(\frac{\partial z}{\partial y} \right)^2 + \frac{z}{c^2} \frac{\partial^2 z}{\partial y^2} = 0 \quad \dots (5)$

From (2), we can write

$$\frac{c^2}{a^2} = -\frac{z}{x} \frac{\partial z}{\partial x} \quad \dots (6)$$

Now multiplying both sides of equation (4) by c^2 , we get

$$\begin{aligned} & \frac{c^2}{a^2} + \left(\frac{\partial z}{\partial x} \right)^2 + z \frac{\partial^2 z}{\partial x^2} = 0 \\ \Rightarrow & -\frac{z}{x} \frac{\partial z}{\partial x} + \left(\frac{\partial z}{\partial x} \right)^2 + z \frac{\partial^2 z}{\partial x^2} = 0 \quad [\text{by using (6)}] \\ \Rightarrow & zx \frac{\partial^2 z}{\partial x^2} + x \left(\frac{\partial z}{\partial x} \right)^2 - z \frac{\partial z}{\partial x} = 0 \quad \dots (7) \end{aligned}$$

Similarly, from (3) and (5), we obtain

$$zy \frac{\partial^2 z}{\partial y^2} + y \left(\frac{\partial z}{\partial y} \right)^2 - z \frac{\partial z}{\partial y} = 0 \quad \dots (8)$$

Here (7) and (8) are two possible forms of the required partial differential equation.

Example-5 Form partial differential equations by eliminating arbitrary functions from the following relations.

(i) $z = f(xy) + g\left(\frac{x}{y}\right)$ **[NUH- 2007]**

(ii) $z = f(x^2 - y) + g(x^2 + y)$ **[NUH- 2007]**

(iii) $y = f(x - at) + \phi(x + at)$ **[NUH- 2006]**

Solution :

(i) Given, $z = f(xy) + g\left(\frac{x}{y}\right)$... (1)

Differentiating (1) partially with respect to x and y respectively, we get

$$\frac{\partial z}{\partial x} = yf'(xy) + \frac{1}{y}g'\left(\frac{x}{y}\right) \quad \dots (2)$$

$$\text{and } \frac{\partial z}{\partial y} = xf'(xy) - \frac{x}{y^2}g'\left(\frac{x}{y}\right) \quad \dots (3)$$

Again, differentiating (2) and (3) partially with respect to x and y respectively, we get

$$\frac{\partial^2 z}{\partial x^2} = y^2 f''(xy) + \frac{1}{y^2}g''\left(\frac{x}{y}\right) \quad \dots (4)$$

$$\text{and } \frac{\partial^2 z}{\partial y^2} = x^2 f''(xy) + \frac{x^2}{y^4}g''\left(\frac{x}{y}\right) + \frac{2x}{y^3}g'\left(\frac{x}{y}\right) \quad \dots (5)$$

$$\begin{aligned} \text{Now, } (4)x^2 - (5)y^2 &\Rightarrow x^2 \frac{\partial^2 z}{\partial x^2} - y^2 \frac{\partial^2 z}{\partial y^2} = x^2 y^2 f''(xy) + \frac{x^2}{y^2}g''\left(\frac{x}{y}\right) - x^2 y^2 f''(xy) - \frac{x^2}{y^2}g''\left(\frac{x}{y}\right) - \frac{2x}{y}g'\left(\frac{x}{y}\right) \\ &\Rightarrow x^2 \frac{\partial^2 z}{\partial x^2} - y^2 \frac{\partial^2 z}{\partial y^2} = -\frac{2x}{y^2}g'\left(\frac{x}{y}\right) \quad \dots (6) \end{aligned}$$

$$\text{Again, } (2)x - (3)y \Rightarrow x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y} = \frac{2x}{y}g'\left(\frac{x}{y}\right) \quad \dots (7)$$

$$\text{Again, } (6) + (7) \Rightarrow x^2 \frac{\partial^2 z}{\partial x^2} - y^2 \frac{\partial^2 z}{\partial y^2} + x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y} = 0$$

which is the desired partial differential equation of 2nd order.

$$(ii) \text{ Given } z = f(x^2 - y) + g(x^2 + y) \quad \dots (1)$$

Differentiating (1) partially w. r. to x , we get

$$\frac{\partial z}{\partial x} = 2xf'(x^2 - y) + 2xg'(x^2 + y)$$

$$\therefore \frac{\partial^2 z}{\partial x^2} = 4x^2 f''(x^2 - y) + 2f'(x^2 - y) + 4x^2 g''(x^2 + y) + 2g'(x^2 + y) \quad \dots (2)$$

Again, differentiating (1) partially w. r. to y , we get

$$\frac{\partial z}{\partial y} = -f'(x^2 - y) + g'(x^2 + y)$$

$$\therefore \frac{\partial^2 z}{\partial y^2} = f''(x^2 - y) + g''(x^2 + y) \quad \dots (3)$$

$$\begin{aligned} (2).x - (3).4x^3 &\Rightarrow x \frac{\partial^2 z}{\partial x^2} - 4x^3 \frac{\partial^2 z}{\partial y^2} = 2xf'(x^2 - y) + 2xg'(x^2 + y) \\ &\Rightarrow x \frac{\partial^2 z}{\partial x^2} - 4x^3 \frac{\partial^2 z}{\partial y^2} = \frac{\partial z}{\partial x} \end{aligned}$$

Which is the required partial differential equation.

$$(iii) \text{ Given } y = f(x - at) + \phi(x + at) \quad \dots (1)$$

Differentiating (1) partially w. r. to x successively twice, we get

$$\frac{\partial y}{\partial x} = f'(x - at) + \phi'(x + at)$$

$$\text{and } \frac{\partial^2 y}{\partial x^2} = f''(x - at) + \phi''(x + at) \quad \dots (1)$$

Again, differentiating (1) partially w. r. to y successively twice, we get

$$\frac{\partial y}{\partial t} = -af'(x-at) + a\phi'(x+at)$$

and
$$\frac{\partial^2 y}{\partial t^2} = a^2[f''(x-at) + \phi''(x+at)]$$

$$\Rightarrow \frac{\partial^2 y}{\partial t^2} = a^2 \frac{\partial^2 y}{\partial x^2} \quad \text{[by using (1)]}$$

which is the required partial differential equation.

Exercise-1

1. Form partial differential equations by eliminating arbitrary constants from the following equations.

(i) $z = zx + (1-a)y + b$

(ii) $z = axy + b$

(iii) $z = ax^3 + by^3$

(iv) $z = a(x+y) + b$

2. Form partial differential equations by eliminating arbitrary functions from the following relations.

(i) $z = e^{mx}\phi(x+y)$

(ii) $z = f(x^2 - y^2)$

(iii) $z = x^n f\left(\frac{y}{x}\right)$

(iv) $z = x + y + f(xy)$

(v) $\phi(x^2 + y^2, z - xy) = 0$